

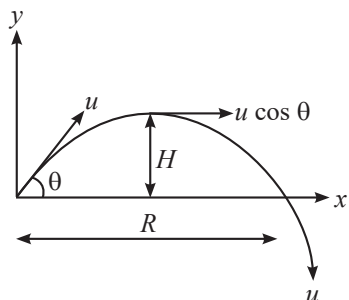
Projection Motion

Horizontal Motion of Projectile

$$u \cos \theta = u_x$$

$$a_x = 0$$

$$x = u_x t = (u \cos \theta) t$$



Vertical Motion of Projectile

$$v_y = u_y - gt \text{ where } u_y = u \sin \theta; y = u_y t - \frac{1}{2} gt^2 = u \sin \theta t - \frac{1}{2} gt^2$$

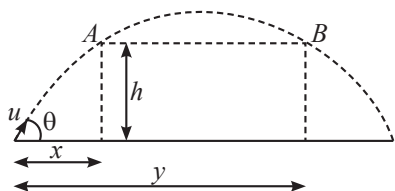
$$\text{Net acceleration} = \vec{a} = a_x \hat{i} + a_y \hat{j} = -g \hat{j}$$

At any Instant

$$v_x = u \cos \theta, \quad v_y = u \sin \theta - gt$$

Ground to Ground Projectile Motion

A body crosses two points at same height in time t_1 and t_2 the points are at distance x and y from starting point then



$$(a) \quad x + y = R$$

$$(b) \quad t_1 + t_2 = T$$

$$(c) \quad h = \frac{1}{2} g t_1 t_2$$

$$(d) \quad \text{Average velocity from A to B is } u \cos \theta$$

If a person can throw a ball to a maximum distance 'x' then the maximum height to which he can throw the ball will be $(x/2)$.

Velocity of Particle at Time t:

$$\vec{v} = v_x \hat{i} + v_y \hat{j} = u_x \hat{i} + (u_y - gt) \hat{j} = u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j}$$

If angle of velocity $\vec{v}$ with horizontal is α , then

$$\tan \alpha = \frac{v_y}{v_x} = \frac{u_y - gt}{u_x} = \frac{u \sin \theta - gt}{u \cos \theta} = \tan \theta - \frac{gt}{u \cos \theta}$$

At highest point: $v_y = 0, v_x = u \cos \theta$

$$\text{Time of flight: } T = \frac{2u_y}{g} = \frac{2u \sin \theta}{g}$$

$$\text{Horizontal range: } R = (u \cos \theta) T$$

$$= \frac{2u^2 \sin \theta \cos \theta}{g} = \frac{u^2 \sin 2\theta}{g} = \frac{2u_x u_y}{g}$$

It is same for θ and $(90^\circ - \theta)$ and maximum for $\theta = 45^\circ$.

$$\text{Maximum height: } H = \frac{u_y^2}{2g} = \frac{u^2 \sin^2 \theta}{2g} = \frac{1}{8} g T^2$$

$$\frac{H}{R} = \frac{1}{4} \tan \theta$$

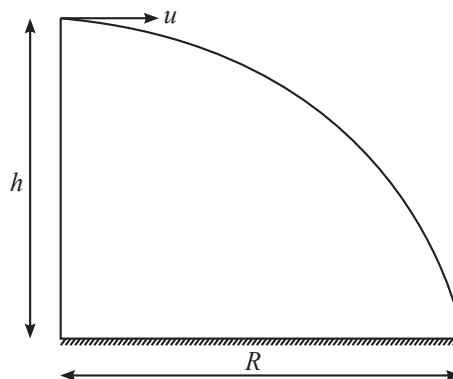
$$\text{Equation of trajectory } y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} = x \tan \theta \left(1 - \frac{x}{R} \right)$$

Horizontal Projection from a Height h

$$\text{Time of flight } T = \sqrt{\frac{2h}{g}}$$

$$\text{Horizontal range } R = uT = u \sqrt{\frac{2h}{g}}$$

$$\text{Angle of velocity at any instant with horizontal } \theta = \tan^{-1} \left(\frac{gt}{u} \right)$$



Projected from Some Height at Some Angle

Case-I: When projected at some angle θ with the horizontal towards upward direction.

time t_1 (time of flight) to strike the ground

$$\therefore t_1 = \frac{T + \sqrt{T^2 + 8h/g}}{2}$$

$$\left(\text{where } T = \frac{2u \sin \theta}{g} \right)$$

$$R = \Delta x = u_x t_1$$

$$v_x = u \cos \theta$$

Case-II: When projected at angle θ with horizontal towards downward direction

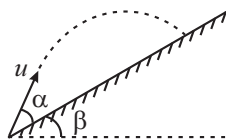
time t_2 to strike the ground then

$$\therefore t_2 = \frac{\sqrt{T^2 + 8h/g} - T}{2}$$

$$R = u_x t_2$$

$$v_x = u \cos \theta$$

Projection on an Inclined Plane



Case-I: Particle is projected up the incline

Time of flight (T):

$$T = \frac{2u \sin \alpha}{g \cos \beta} = \frac{2u_{\perp}}{g_{\perp}}$$

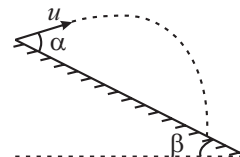
Maximum height from inclined plane (H):

$$H = \frac{u^2 \sin^2 \alpha}{2g \cos \beta} = \frac{u_{\perp}^2}{2g_{\perp}}$$

Range along the inclined plane (R):

$$R = \frac{2u^2 \sin \alpha \cos(\alpha + \beta)}{g \cos^2 \beta}$$

Case-II: Particle is projected down the incline



Time of flight (T):

$$T = \frac{2u \sin \alpha}{g \cos \beta} = \frac{2u_{\perp}}{g_{\perp}}$$

Maximum height (H):

$$H = \frac{u^2 \sin^2 \alpha}{2g \cos \beta} = \frac{u_{\perp}^2}{2g_{\perp}}$$

Range along the inclined plane (R):

$$R = \frac{2u^2 \sin \alpha \cos(\alpha - \beta)}{g \cos^2 \beta}$$